Spin Glasses and Neural Networks: Multilayer Sherrington–Kirkpatrick Model

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Motivation

Analogy between Neural Networks and Spin Glasses:

- Both systems seek out stable configurations by minimizing a cost functional or energy.

Physical perspective:

- Statistical-physics tools (e.g. replica analysis, TAP equations) can describe their evolution and assess stability.

Motivation:

Leverage rigorous physical methods to deepen our understanding of complex ML landscapes

What Is a Spin Glass?

A magnetic system where spins interact through random, competing couplings, preventing uniform order—even at low temperature.

Key features :

- Exponentially many local energy minima
- Mixture of short- and long-range order
- Exponential number of metastable states

Interaction types:

- Positive (ferromagnetic)
- Negative (antiferromagnetic)

The Sherrington–Kirkpatrick Model (MSK)

An Ising spin system with an infinite radius of interaction - each spin interacts with all the others with a random coefficient

$$H_{MSK} = -\sum_{i<j} J_{ij}\,\sigma_i\sigma_j, \quad J_{ij} \sim \mathcal{N}\left(0, \frac{1}{N}\right)$$ [1]

Paramagnetic phase:

- At high T, spins are uncorrelated and flip rapidly—system behaves chaotically

Freezing transition:

- the system becomes non-ergodic, trapping into metastable states

Replica-symmetric solution:

Loses stability for T<Tc, leading to negative entropy at zero temperature

Parisi Ansatz → **RSB**

1)D. Sherrington and S. Kirkpatrick, “Solvable model of a spin-glass”, Phys. Rev. Lett. 35,1792–1796 (1975).

Neural Network as Multilayer Spin Model

Edwards–Anderson (EA): local lattice couplings

(EA) + (MSK) = (MCM)

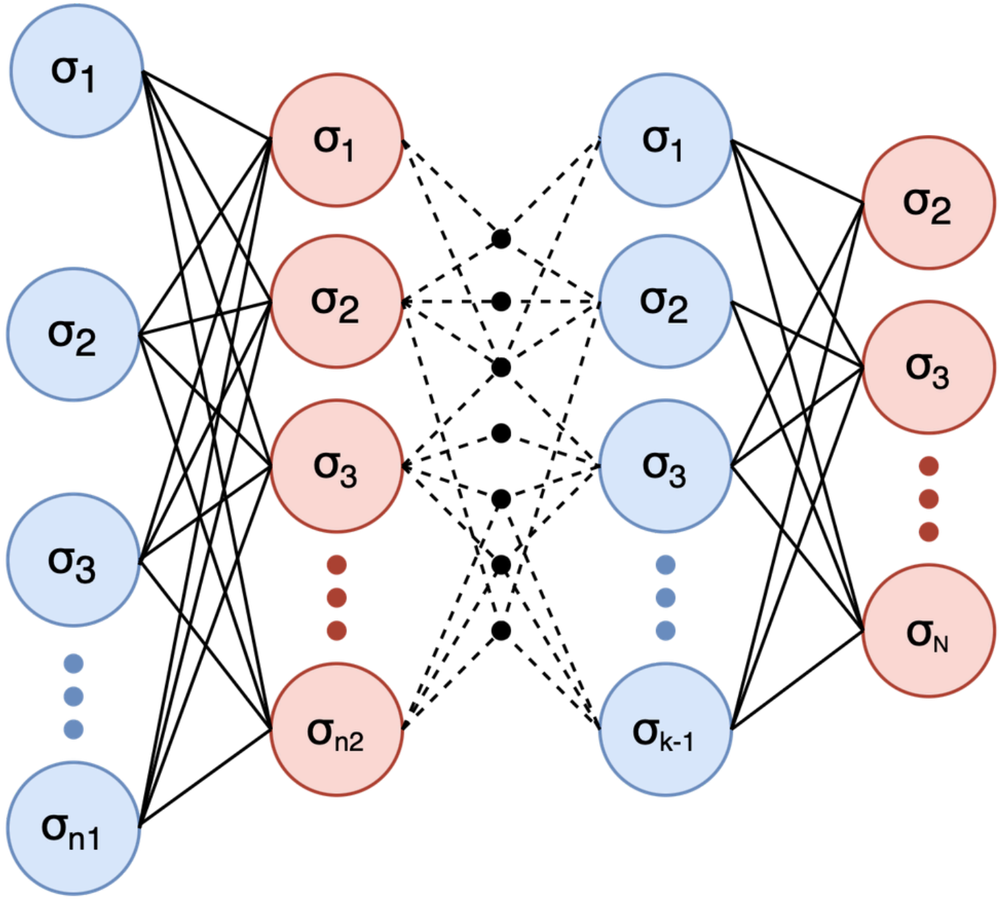
Neuron as spin $\sigma_i = \pm 1$

Weight as coupling $W_{ij}^{(k)} \to J_{ij}^{(k)}$

Hamiltonian (Multi-Layer Spin Model, MSM):

$$H_{\mathrm{MSK}} = -\sum_{k=1}^{L}\sum_{i=1}^{N_{k-1}}\sum_{j=1}^{N_k} J_{ij}^{(k)}\,\sigma_i^{(k-1)}\,\sigma_j^{(k)}$$

$$J = \begin{pmatrix} 0 & J^{(1)} & 0 & \cdots & 0 \\ (J^{(1)})^T & 0 & J^{(2)} & \cdots & 0 \\ 0 & (J^{(2)})^T & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & J^{(L)} \\ 0 & 0 & 0 & (J^{(L)})^T & 0 \end{pmatrix}$$



Simulated Annealing in Spin Systems

Simulation of the dynamic process of spin flips using the **Monte Carlo** method:

1. Start in a high-temperature, random state.
2. Propose random spin flips (Monte Carlo)
3. Compute energy change ΔE.
4. **The Metropolis Criterion:**
    - If ΔE < 0 — accept
    - If ΔE > 0 — accept with probability: $e^{-\frac{\Delta E}{kT}}$
5. Gradually lower temperature
6. System "freezes" into one of the energy minima

Visualizing Annealing

- **High T (Paramagnet):**

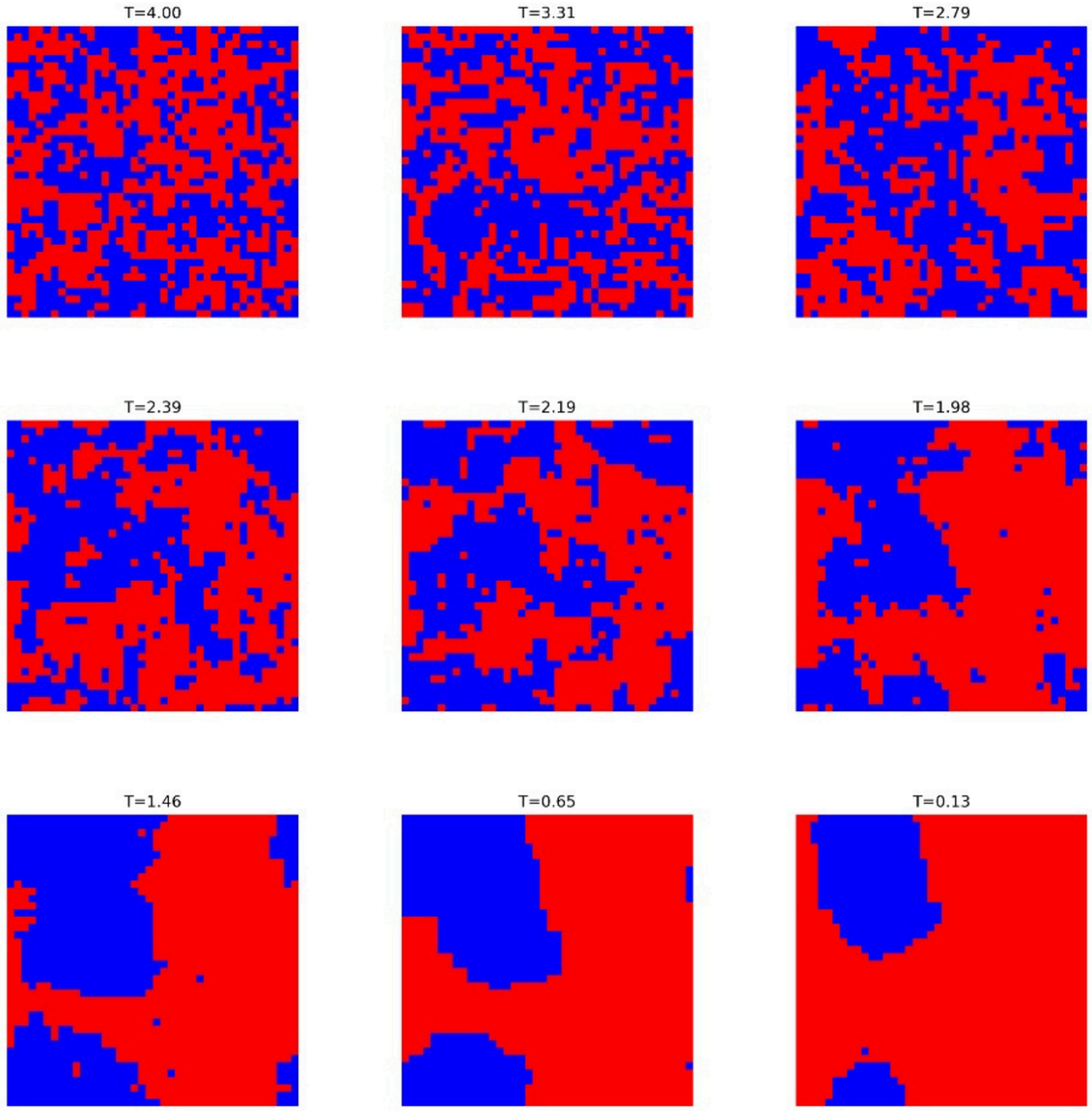
Rapid, random flips; no long-range order.

- **Phase transition:**

Emergence of stable clusters as T drops.

- **Low T:**

Configuration stabilizes into a glassy or ordered state.



Training of system

Stochastic Gradient Descent minimizes a loss function → gradually improves predictions.

- **Initial state:** Weights random ⇒ analogous to paramagnetic spins.
- • **During training:** Loss decreases ⇒ “energy” of the spin model lowers, leading to a more stable configuration.
- • **Effectively:** Cooling allows the network to settle into a metastable, task-relevant state.

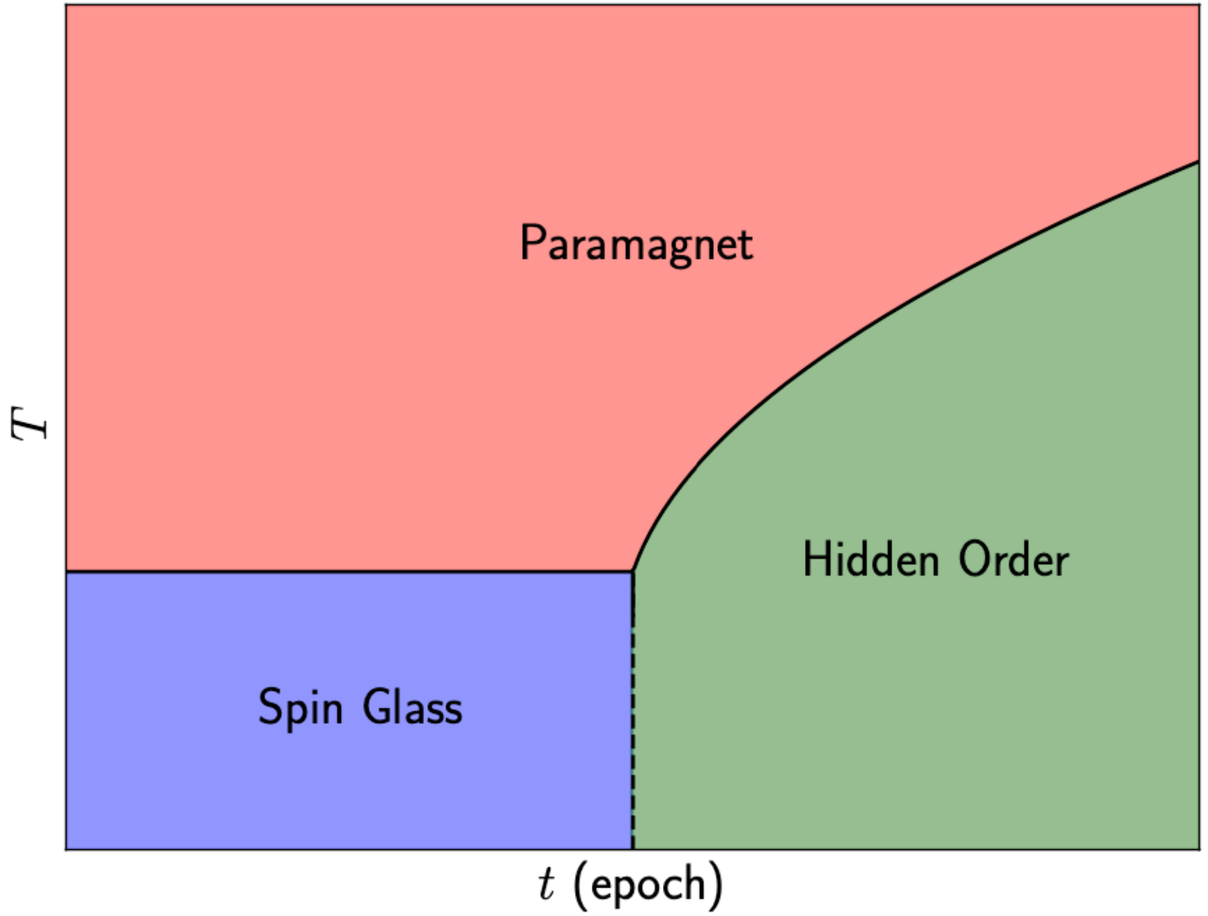


График с D. J. Thouless, P. W. Anderson, and R. G. Palmer, “Solution of ‘solvable model of a spin glass’”, Philosophical Magazine 35, 593–601 (1977).

Recipe for the Binarized Network

L2-hinge loss :

$$L = \frac{1}{N} \sum_{i=1}^{N} \sum_{j \neq y_i} \max\left(0, 1 - \hat{y}_{y_i} + \hat{y}_j\right)^2$$

SGD with momentum:

1. **Initialization**: Random weights (Kaiming normal).
2. **Forward pass**: Binary activations.
3. **Backward pass**: Straight-Through Estimator (STE) for gradients through sign: surrogate hard-tanh.
4. **Weight update**: SGD with momentum
5. **Iterate** until convergence.

Optionally combine with simulated-annealing-style temperature scheduling

Straight-Through Estimator (STE)

Activation is non-differentiable ⇒ zero gradient almost everywhere.

STE - method of direct approximation of the gradient through a surrogate continuous function

$$\varphi_{\text{forward}}(x) = \text{sign}(x) \xrightarrow{\textbf{STE}} \varphi_{\text{back}}(x) = \text{htanh}(x) = \begin{cases} -1, & x < -1 \\ x, & -1 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

- **Forward pass:** use hard sign
- **Backward pass:** pretend derivative is that of hard-tanh surrogate

$$\delta^{(l)} = (\delta^{(l+1)} \cdot W^{(l+1)}) \cdot \varphi\prime(z^{(l)})$$

$$\frac{\partial L}{\partial W_{ij}} = a_i \cdot \delta_j \qquad v_t = \mu\, v_{t-1} - \eta \nabla L, \quad W \to W + v_t$$

Critical Temperature & Replica/TAP Analysis

Replica method (MSK)[1]:

Ensemble-average yields critical Tc via stability of the replica-symmetric solution

$$T_c = \left[2C \cos \frac{\pi}{L+2}\right]^{-1/2} \quad \Longrightarrow \quad T_c \approx \sqrt{2C}, L \to \infty.$$

TAP (Thouless-Anderson-Palmer)[2]:

Free energy:

$$F[m] \;=\; -\frac{1}{2}\sum_{ij} J_{ij}\, m_i m_j \;-\; \frac{\beta}{4}\sum_{ij} J_{ij}^2 (1-m_i^2)(1-m_j^2) \;+\; \frac{T}{2}\sum_i \left\{(1+m_i)\ln\frac{1+m_i}{2} + (1-m_i)\ln\frac{1-m_i}{2}\right\}$$

- In a real trained network, the weights J_{ij} are fixed
- It is necessary to analyze the energy landscape for a specific set

$$m_i = \langle \sigma_i \rangle \quad \overset{1}{\Longrightarrow} \quad \tanh^{-1}(m_i) = \beta \sum_j J_{ij}\, m_j - \beta^2\, m_i \sum_j J_{ij}^2\,(1-m_j^2)$$

1) Barney R., Winer M., Galitski V. Neural Networks as Spin Models: From Glass to Hidden Order Through Training
2) D. J. Thouless, P. W. Anderson, and R. G. Palmer, “Solution of ‘solvable model of a spin glass’”, Philosophical Magazine 35, 593–601 (1977).

Calculation of the critical temperature

Linearization near $m_i \approx 0$:

$$m_i \approx \beta \sum_j J_{ij} m_j - \beta^2 \sum_j J_{ij}^2 m_i$$

The solution of this system provides a criterion for the appearance of a non–trivial solution - determines Tc

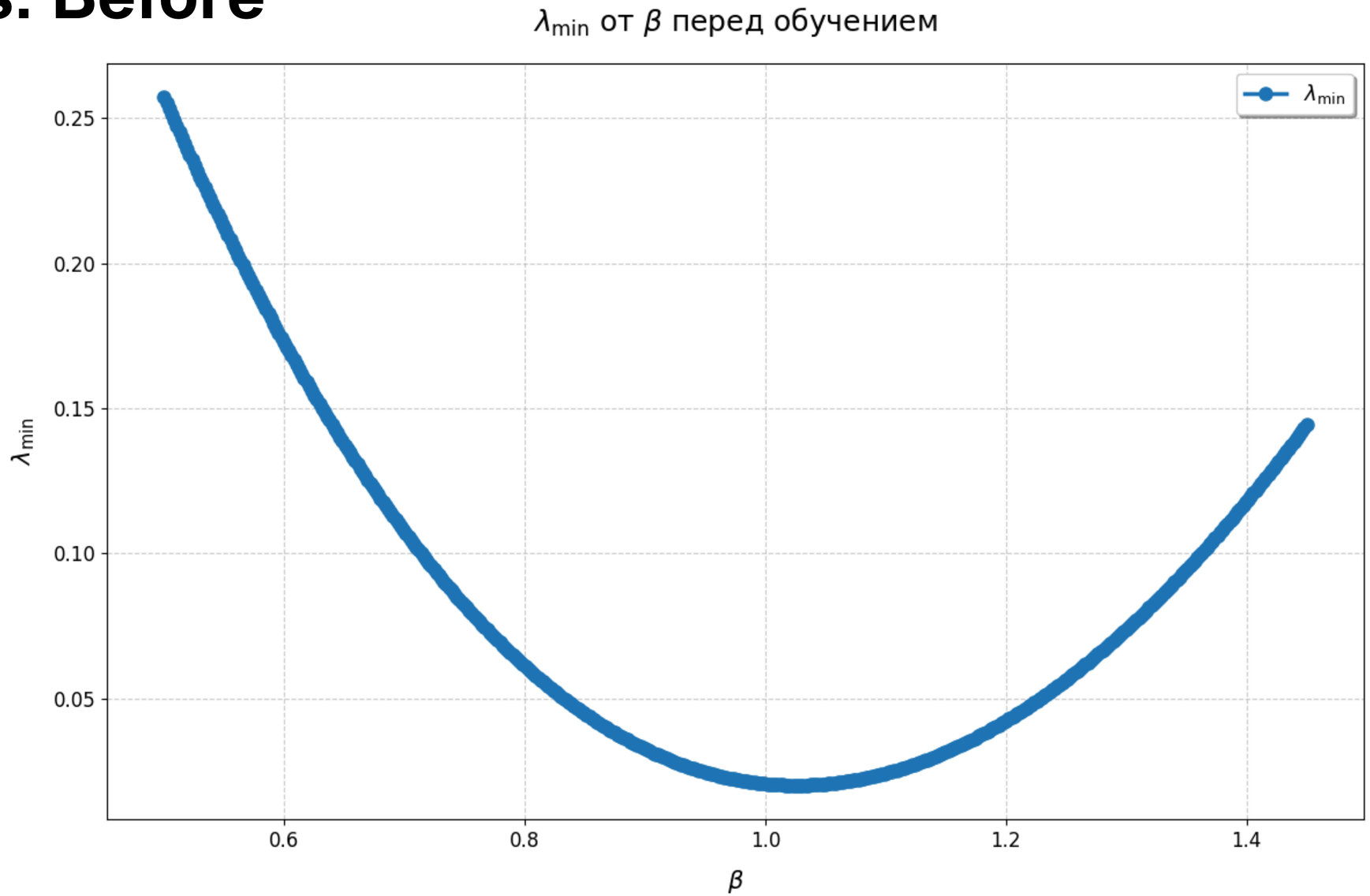
We are looking for the minimum eigenvalue of the matrix:

$$(I_N - M)m = 0, \quad M_{ij} = \beta J_{ij} - \delta_{ij}\beta^2 \sum_\ell J_{i\ell}^2$$

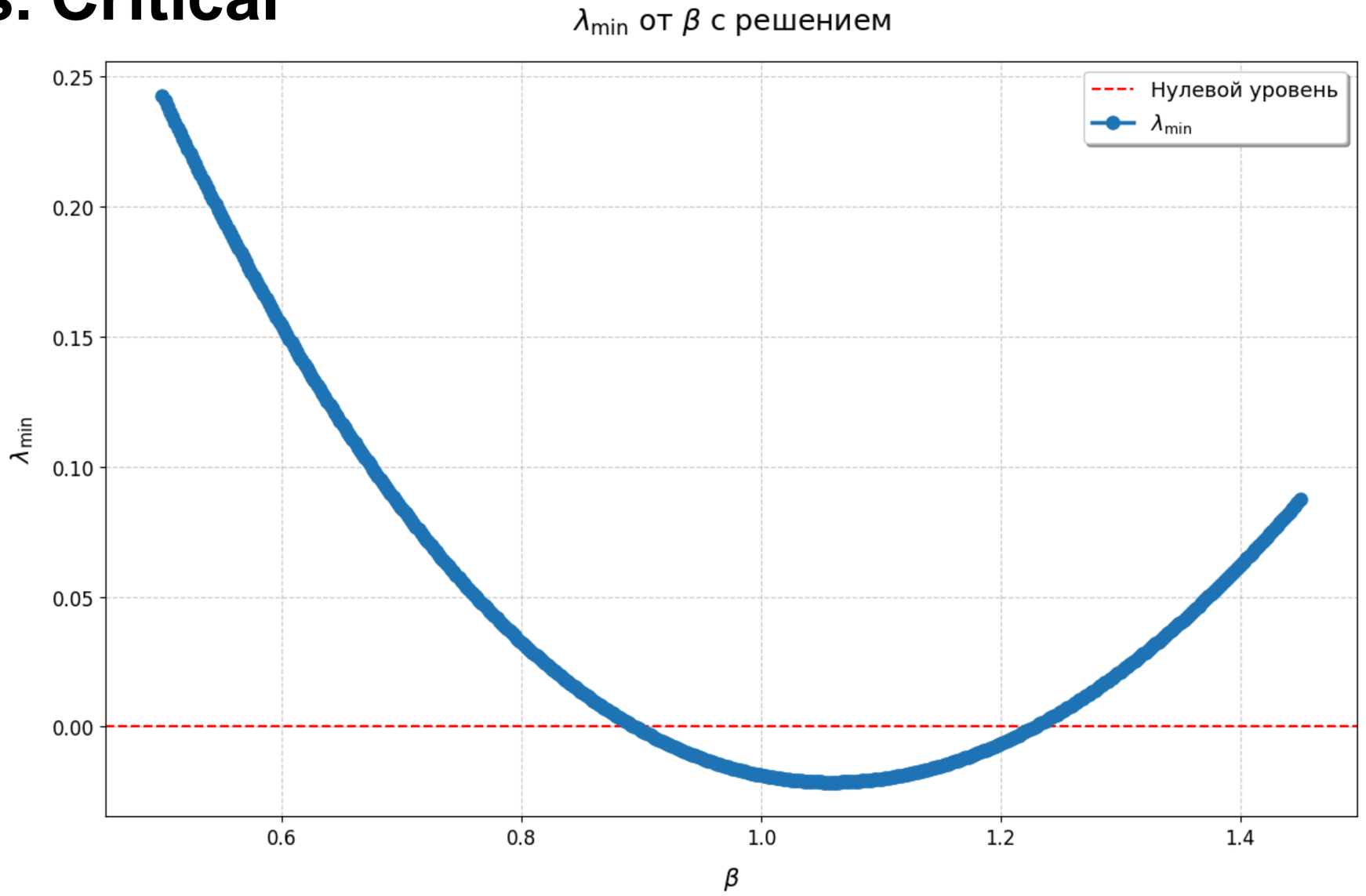
Algorithm for Tc:

1. Form a matrix of weights
2. Calculating the minimum eigenvalue $\lambda_{\min}$
3. The critical temperature corresponds to β_c , at which $\lambda_{\min}$ becomes zero: $T_c = \frac{1}{\beta_c}$

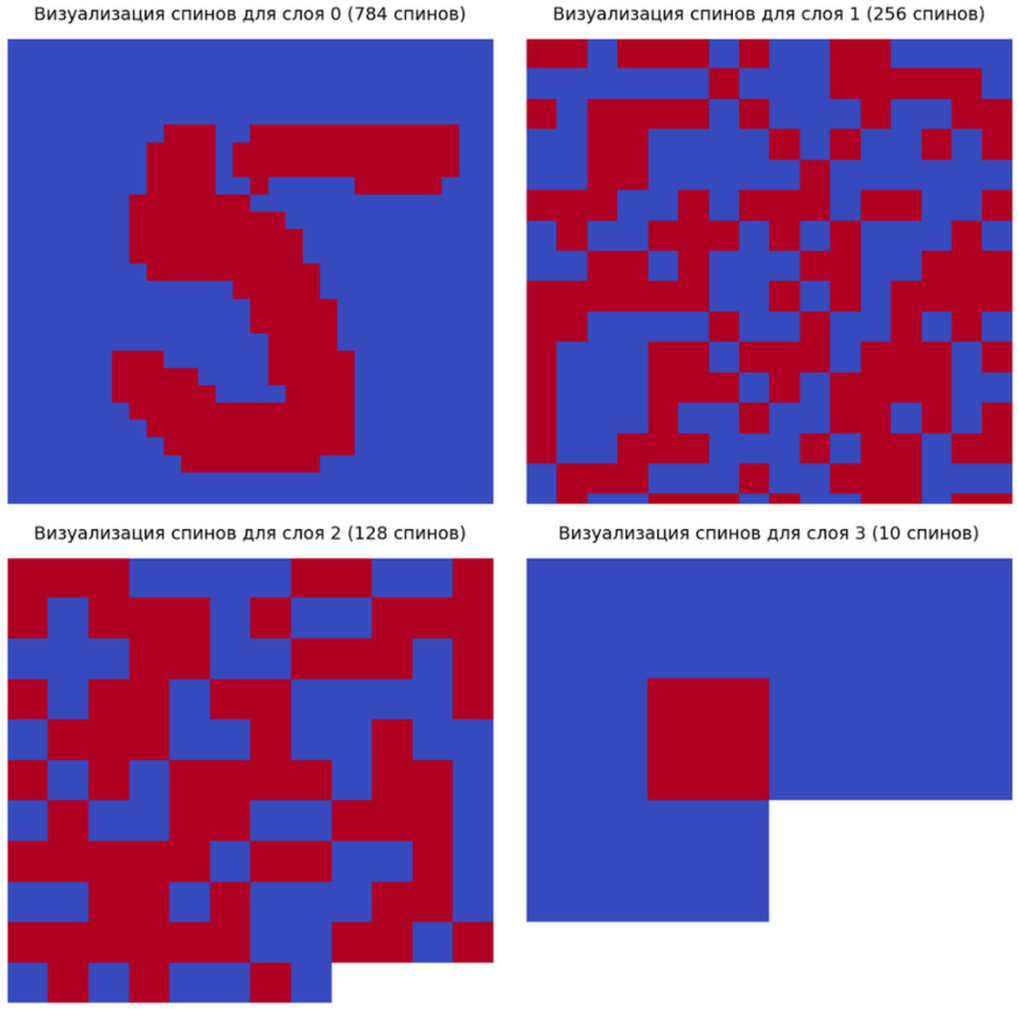
Results: Before



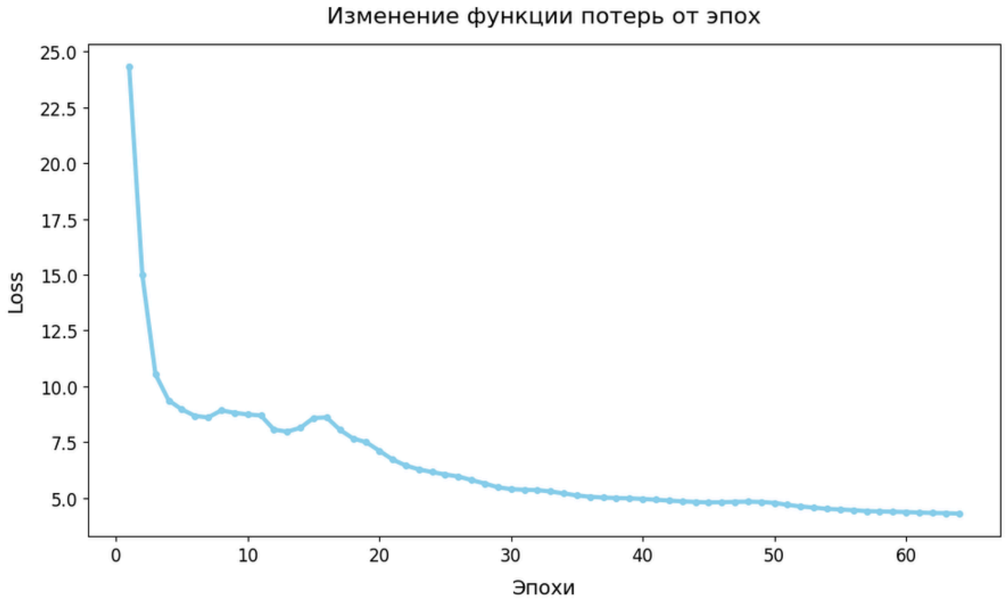
Results: Critical

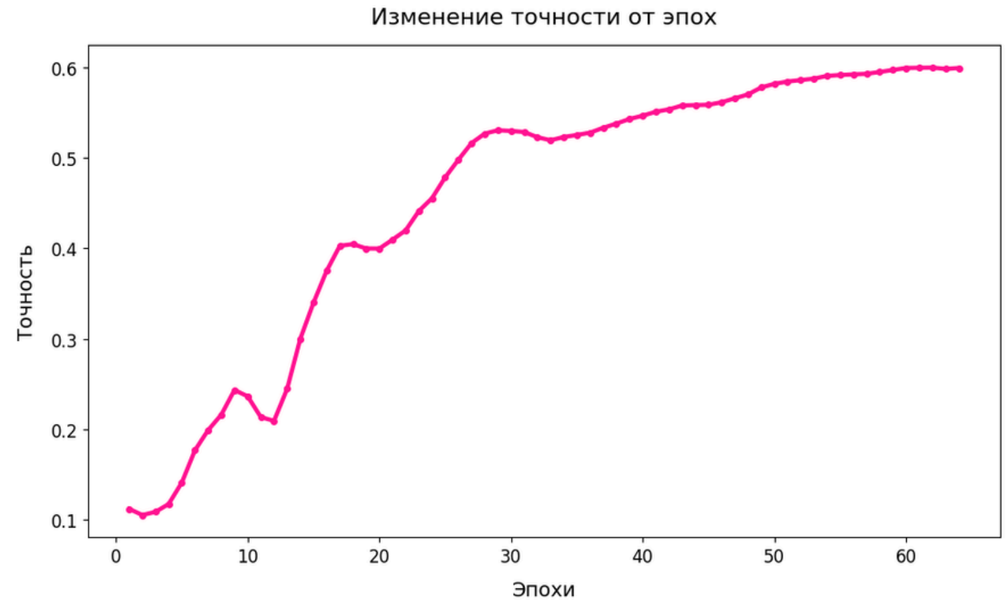


Results: training

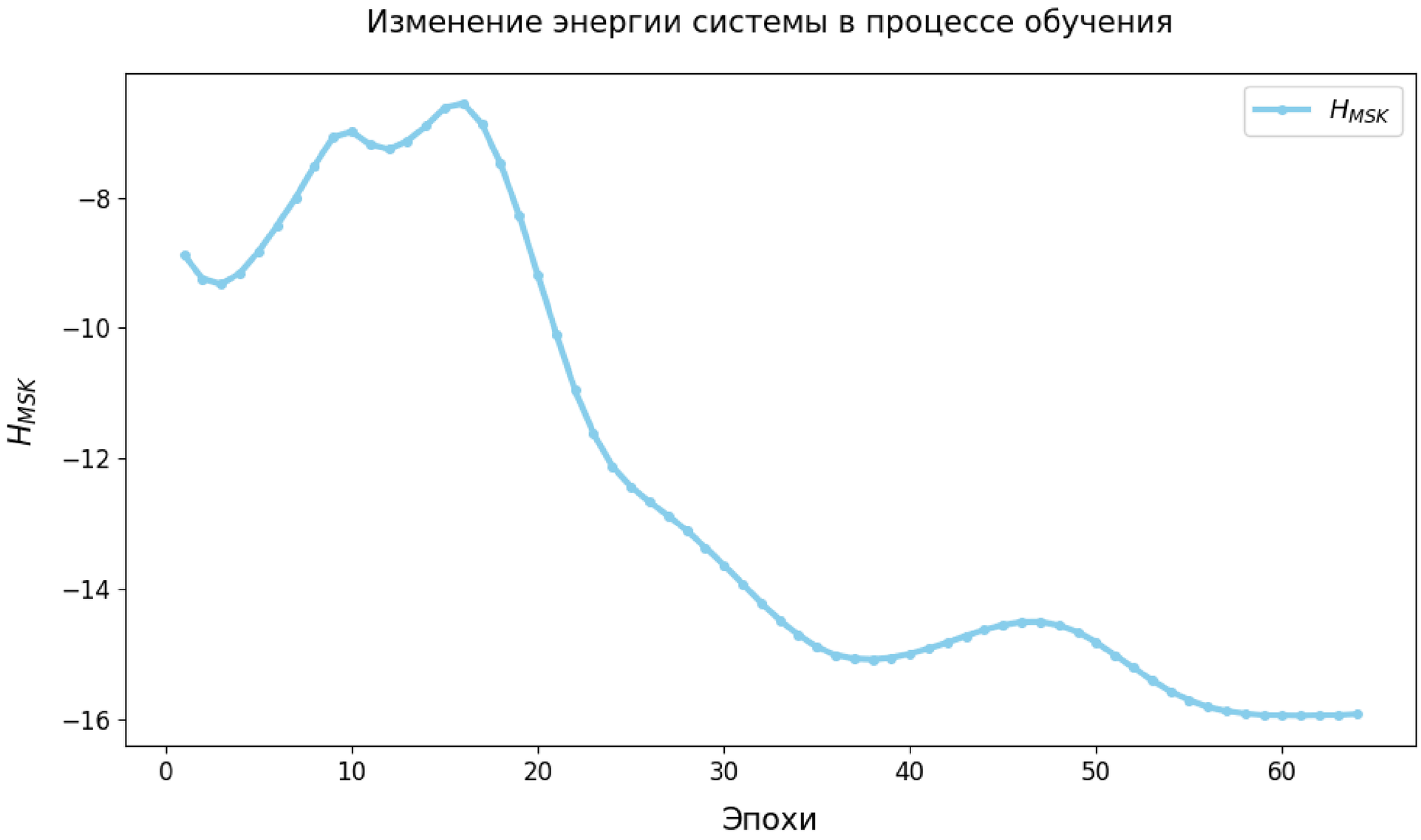


MNIST dataset

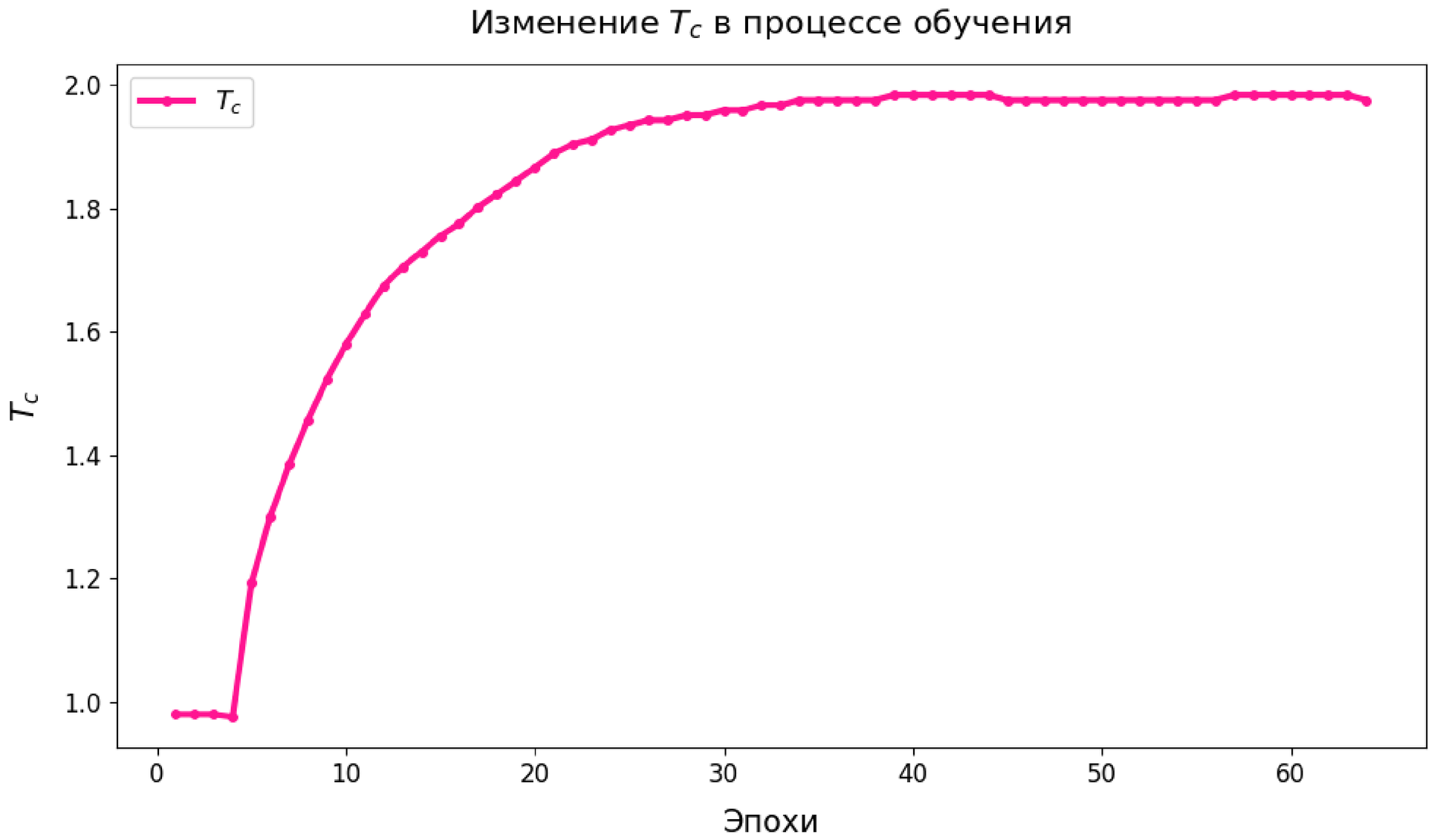




Results: training



Results: training



Conclusions

- A randomly initialized network is formally **equivalent** to a layered SK spin glass with replica-symmetry breaking and a well-defined critical temperature Tc.
- Replica-method stability analysis and the **TAP** equations accurately predict Tc and identify the glass-to-ordered phase transition in the loss landscape.
- **SGD with momentum** and **STE** performs a gradual “cooling,” driving the system out of its initial glassy state into a single symmetry-broken configuration (*hidden order*).
- Throughout training, the effective Tc grows **monotonically**—signaling that the learned solution resides in ever-deeper, more robust minima less susceptible to perturbations.
- Viewing training through the lens of spin-glass physics opens new diagnostic tools (eigenvalue spectra, TAP-based stability checks) and suggests **physics-inspired regularizers** and annealing schedules to improve optimization and generalization in deep networks.

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